Student Performance Analytics System

MINI PROJECT REPORT

Submitted by

HARIPRASATH M 21172400700109

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IN

ARTIFICIAL INTELLIGENCE AND DATA SCIENCE

RAJALAKSHMI INSTITUTE OF TECHNOLOGY

KUTHAMBAKKAM, CHENNAI - 600 124

ANNA UNIVERSITY: CHENNAI 600 025

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ANNA UNIVERSITY: CHENNAI - 600 025

BONAFIDE CERTIFICATE

This is to Certified that this Mini Project report ***"Student Performance Analytics System"*** is the Bonafide work of ***"HARIPRASATH M (2117240070109)"***, who carried out the project work under my supervision.

SIGNATURE

HEAD OF THE DEPARTMENT

Dr. N. KANAGAVALLI, Ph.D

Assistant Professor,

AI & DS DEPARTMENT

Rajalakshmi Institute of Technology,
Chennai – 600 124.

SIGNATURE

SUPERVISOR

Mrs. RUBINA BEGAM

Assistant Professor,

AI & DS DEPARTMENT

Rajalakshmi Institute of Technology,
Chennai – 600 124.

Submitted for the Mini Project viva-voce held on ……..

INTERNAL EXAMINER

EXTERNAL EXAMINE

INTRODUCTION

In the field of education, large volumes of student-related data such as attendance, internal marks, assignment scores, and final grades are generated continuously. However, this data is often underutilized and not properly analyzed to extract meaningful insights.

Analyzing student performance data helps in understanding the factors that influence academic success. It allows educators to identify patterns, trends, and relationships between different academic parameters.

This project, titled "Student Performance Analytics System," focuses on analyzing student data using statistical and visualization techniques. The system processes various academic attributes and identifies how each factor contributes to student performance.

By applying descriptive statistics and visualization methods, the project provides insights into student behavior and performance trends. This helps in improving teaching strategies and student outcomes.

PROBLEM STATEMENT

Educational institutions collect large amounts of student data, but analyzing this data manually is difficult and inefficient.

There is a lack of proper tools to identify key factors affecting student performance, understand relationships between academic parameters and make data-driven decisions.

Hence, there is a need for a system that can analyze student data using statistical methods and visualization techniques to derive meaningful insights.

GOAL

The main goal of this project is to analyze student academic data and identify the major factors influencing student performance using statistical analysis and visualization techniques.

THEORETICAL BACKGROUND

Descriptive Statistics:

Descriptive statistics summarize data using:

- Mean → Average value

- Variance → Spread of data

- Standard Deviation → Data consistency

Correlation Analysis:

Correlation measures the relationship between variables:

- Positive correlation → both increase together
- Negative correlation → one increases, other decreases

Data Visualization:

Visualization helps in understanding data patterns:

- Scatter Plot → relationship between variables
- Box Plot → distribution and outliers

- Histogram → frequency distribution

Tools and Technologies:

- Python
- Pandas
- NumPy
- Matplotlib
- Seaborn

ALGORITHM EXPLANATION

Step 1 - Data Collection:

Collect dataset containing student details like attendance, marks, assignments and grades.

Step 2 - Data Loading:

Load dataset using Pandas

df = pd.read_csv("studentperformance.csv")

Step 3 – Data Cleaning:

Handle missing values and format data.

Step 4 - Statistical Analysis:

Calculate mean, variance, and correlation.

Step 5 - Visualization:

Generate scatter plots, box plots, and histograms.

Step 6 - Interpretation:

Analyze graphs to identify important factors.

Step 7 – Conclusion:

Determine key factors affecting performance.

IMPLEMENTATION AND CODE

```
from flask import Flask, render_template, request
import pandas as pd
import plotly.express as px
from pathlib import Path

app = Flask(__name__)

def get_dataset():
    for f in Path(".").glob("*"):
        if f.suffix in [".csv", ".xlsx", ".xls"]:
            return f
    return None

def load(file):
    return pd.read_csv(file) if str(file).endswith(".csv") else pd.read_excel(file)

def analyze(df):
    df_num = df.select_dtypes(include="number")
    stats = df_num.describe().T
    corr = df_num.corr().round(2)

    hist = px.histogram(df_num).to_html(full_html=False)
    scatter = None
    if len(df_num.columns) > 1:
        cols = df_num.columns
        scatter = px.scatter(df_num, x=cols[0], y=cols[1]).to_html(full_html=False)
```

```
    return {
        "stats": stats.to_html(classes="table"),
        "corr": corr.to_html(classes="table"),
        "hist": hist,
        "scatter": scatter,
        "preview": df.head().to_html(classes="table")
    }

@app.route("/", methods=["GET", "POST"])
def index():
    df = None
    if request.method == "POST" and "dataset" in request.files:
        file = request.files["dataset"]
        if file.filename:
            df = load(file)
    if df is None:
        path = get_dataset()
        if path:
            df = load(path)

    results = analyze(df) if df is not None else None
    return render_template("index.html", results=results)

if __name__ == "__main__":
    app.run(debug=True)
```

OUTPUT:

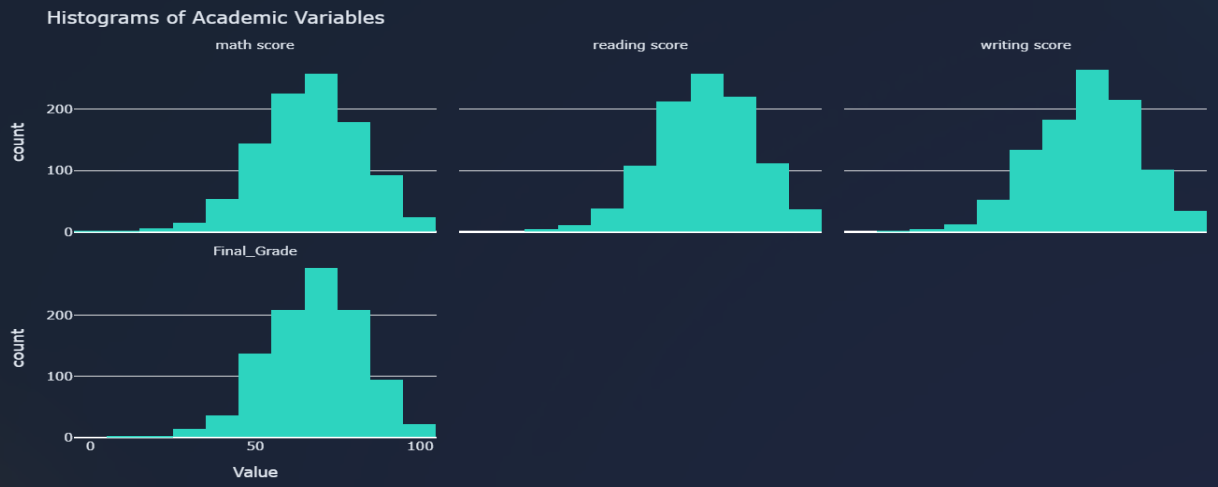
Descriptive Statistics

variable	mean	median	variance	std_dev	min	max
math score	66.089	66.0	229.919	15.163	0.0	100.0
reading score	69.169	70.0	213.166	14.600	17.0	100.0
writing score	68.054	69.0	230.908	15.196	10.0	100.0
Final_Grade	67.770	68.3	203.268	14.257	9.0	100.0

Correlation Matrix

	math score	reading score	writing score	Final_Grade
math score	1.000	0.818	0.803	0.919
reading score	0.818	1.000	0.955	0.970
writing score	0.803	0.955	1.000	0.966
Final_Grade	0.919	0.970	0.966	1.000

Histograms



Bar Chart

Scatter Plots vs Performance Target

RESULTS AND FUTURE ENHANCEMENT

Result:

The system successfully analyzes student data and identifies the factors influencing academic performance. It was observed that:

- Attendance has a strong impact on final marks
- Assignment scores contribute significantly
- nternal marks show positive correlation with final grades

Comparison with Other Approaches:

Traditional methods rely on manual analysis, which is time-consuming. This system provides faster and more accurate insights using data analytics.

Future Enhancements:

1. Use real-time student data
2. Apply machine learning models for prediction
3. Develop a web dashboard
4. Integrate with college ERP systems

Git Hub Link for the project and report

GitHublink of the project and report	hariprasath729/DA-mini-project

REFERENCES:

1. Python Documentation
2. Pandas Documentation
3. Matplotlib Documentation
4. Seaborn Documentation
5. Kaggle Dataset